

SLM vs LLM

Why smaller Gen AI models are better

—By **Jayesh Shinde**

In the realm of artificial intelligence, especially Generative AI, we've all been familiarised with the term LLM or Large Language Model for some time now. For the uninitiated, "SLM" might sound unfamiliar, but these Small Learning Models are playing an increasingly vital role in various technological applications.

LLMs like OpenAI's GPT-4 or Meta's Llama 3.1 are trained on vast amounts of data with the end goal of performing a wide range of tasks across various domains. However, as the AI landscape evolves, a new paradigm of specialised artificial intelligence is emerging, represented by Small Language Models (SLMs), according to Dr Aakash Patil, Deep Learning Researcher at Stanford University.

"SLMs are designed to be more compact and efficient, typically containing fewer parameters than LLMs. This smaller size doesn't necessarily mean reduced capability; rather, it often translates to faster processing and lower computational costs, especially in resource-constrained environments," says Dr Patil.

Small Language Models (SLMs) have already started rocking the AI industry in ways that many didn't anticipate, suggests Prem Naraindas, CEO and Founder of Katonic AI. "While the hype has largely centred around behemoth models like GPT-4, I've long maintained that the real revolution will come from more nimble, specialised models," he says.

SLM vs LLM: Bigger isn't always better

There's no fixed definition of how small



an SLM needs to be for it to be called an SLM. At the moment, "small and nimble" is roughly defined as anything under 20 billion parameters, according to Intel's researchers. The size threshold is a moving target that may double later in the coming year, but it does give you an idea how SLMs are significantly smaller in size than the likes of snapshot comparison against the 175 billion parameters of ChatGPT 3.5 or 1.76 trillion parameters size of ChatGPT 4, for instance.

When it comes to Generative AI, developers are inundated with choices – perhaps too many, believes Moshe Berchansky, Senior AI Researcher at Emergent AI Research Labs, Intel. A small number of giant models are good for general and multi-purpose use cases, and a giant number of small models are for added efficiency, accuracy, security, and traceability. If you decide to become an entrepreneur and build your own AI startup, there's plenty of decisions to take between opting for an LLM vs SLM.

Giant models, while versatile, can be computationally expensive and may

not be ideal for resource-constrained environments. Smaller, nimbler models, often 10x-100x smaller, offer improved efficiency and accuracy for targeted tasks. Whether a model is general-purpose or targeted depends on the specific needs of the application. Finally, the decision between cloud-based and local inference depends on factors like latency, privacy, and cost.

"While LLMs require enormous computational resources for training and inference, SLMs demonstrate that relatively less compute can yield more reliable and accurate responses in specialised contexts. This efficiency is particularly crucial for industries where precision and domain-specific knowledge are paramount," says Dr Aakash Patil.

According to Intel's research, Dolly, Stable Diffusion, StarCoder, DALL-E, and Phi are powerful examples of models at this scale. Microsoft Research's Phi 2, at 2.7 billion parameters, recently showcased amazing progress of so-called "small language models" on benchmarks in terms of common sense, language